Lithium Site Selection (LiSelect · ZW)

A decision-support web application for choosing where to site a lithium processing plant in Zimbabwe. It screens candidate sites against hard feasibility gates, scores them against weighted criteria, selects the minimum-cost site while allocating mine feedstock to it, and tests how stable that recommendation is when assumptions change.

---

1. How the application meets the project objectives

Objective 1 — Structured criteria framework with required evidence

> To develop a structured criteria framework that identifies and defines the industrial, logistical, utility, land and environmental factors influencing candidate-site suitability together with the evidence required to assess each factor.

Requirement	Where it is met
Factors grouped as industrial, logistical, utility, land, environmental	**Criteria** page. The 15 criteria are stored in the `Criterion` table, each with a group (`grp`).
Each factor is defined	Every criterion has a `definition`.
Evidence required to assess each factor	Every criterion has `evidence` (what must be collected) and `source` (where it comes from).
Measurement basis	Each criterion has a data field, a direction (benefit or cost), a normalisation range (min–max) and a unit.
Hard feasibility requirements	The Criteria page lists the 8 feasibility gates, each with threshold, definition and required evidence (`lib/gates.ts`).

Objective 2 — Transparent mathematical optimisation model

> To formulate a transparent mathematical optimisation model for facility selection and inbound feedstock allocation incorporating clearly stated capacity, supply and feasibility constraints.

The model is implemented in `lib/evaluate.ts`, `lib/transport.ts` and `lib/gates.ts`, and is printed with live figures in the **Optimisation model** panel of every scenario page.

```
Sets:      i = mines, j = candidate sites
```

```
Data:      S_i = annual supply of mine i, C_j = capacity of site j,
           d_ij = road km, r = haulage rate (USD/t·km),
           F_j = fixed opex, L_j = land cost / 20-year amortisation
Variables: y_j  {0,1} (site j selected), x_ij  0 (t/yr from mine i to site j)

min  Σ_j [ F_j + L_j ] y_j + Σ_i Σ_j r · d_ij · x_ij
s.t. Σ_j y_j = 1                       exactly one facility
     Σ_j x_ij  S_i        i           mine supply limit
     Σ_i x_ij = C_j · y_j   j         plant fed to capacity if selected
     y_j = 0 if site j fails any feasibility gate
```

- **Capacity constraint:** a selected plant must be fed exactly its capacity.
- **Supply constraint:** a mine cannot ship more than its annual supply, and a site is infeasible if total mine supply is below its capacity.
- **Feasibility constraints:** water, grid, settlement buffer, protected land, capacity, land area, tailings suitability and supply sufficiency.
- **Solution method:** every gate-passing site is evaluated. For each one the allocation is solved exactly (ship from the nearest mines first, which is optimal for a single destination with constant unit costs). The site with the lowest total annual cost is selected.
- **Weighted score:** a 0–100 score is also computed for every feasible site. It informs the decision and is reported alongside the cost result, and the scenario page flags when the top-scoring site differs from the cost-optimal one.

Objective 3 — User-facing computational system

> To develop a user-facing computational system that reports feasibility-gate outcomes, estimated annual costs, allocation decisions and sensitivity to changing assumptions.

Each scenario page (`/scenarios/[id]`) reports:

Required output	Section on the scenario page
Feasibility-gate outcomes	**Feasibility gates**: PASS/FAIL per site, with the measured value for each gate.
Estimated annual costs	**Estimated annual costs**: haulage, fixed opex, land amortisation and total, ranked by cost.
Allocation decisions	**Feedstock allocation**: tonnes per mine, road km and cost for the recommended site, plus a map of the haulage legs.

Required output	Section on the scenario page
Sensitivity to assumptions	**Sensitivity**: haulage rate, mine supply and criterion weights, with a robust or fragile verdict.

---

2. Running the application

Requires Node.js and pnpm.

```
pnpm install
npx prisma migrate deploy   # create / update the SQLite database (prisma/dev.db)
npx prisma generate         # generate the Prisma client
pnpm dev                    # http://localhost:3000
```

On first start the Dashboard shows **Load Zimbabwe demo dataset**. Click it to load 5 candidate sites, 5 mines and 15 criteria.

Other commands: `pnpm build`, `pnpm start`, `pnpm lint`.

---

3. Recommended workflow

1. **Mines**: enter the feedstock sources and their annual supply.
2. **Sites**: enter the candidate sites and their measured attributes.
3. **Criteria**: review the definitions and evidence, then adjust the weights.
4. **Dashboard**: set the haulage rate and run a scenario.
5. **Scenario page**: read the recommendation, then check the gates, costs, allocation and sensitivity.

---

4. Pages and inputs

Dashboard (`/`)

Purpose: run scenarios and open past results.

Input	Purpose
Load Zimbabwe demo dataset (only when the database is empty)	Seeds example sites, mines and criteria.
Scenario name	Label for the saved run. Defaults to a timestamp.
Haulage rate (USD/t · km)	Unit cost of road transport, `r` in the model. Default is 0.25.

Input	Purpose
Run selection pipeline	Runs the full analysis and saves the result as a scenario.
Recent scenarios list	Opens or deletes the last 10 runs. Each run is a saved snapshot.

The tiles show how many sites, mines and criteria are loaded and the total criterion weight.

Mines (`/mines`)

Purpose: define the feedstock supply that the plant draws on.

Input	Purpose
Name	Identifies the mine in allocations and on the map.
Latitude / Longitude	Used to compute road distance to each site (straight-line distance × 1.3 road factor).
Annual supply (t)	Supply limit `S_i`. A mine never ships more than this, and total supply must cover a site's capacity.
Notes	Free text, such as ore type. It is not used in calculations.

Sites (`/sites`)

Purpose: define candidate plant locations. The table shows a live PASS or FAIL on the site-only gates. The supply gate is added when a scenario runs.

Input	Purpose
Name, Latitude, Longitude	Identification, map position and distance to mines.
Capacity (t/yr)	Plant throughput `C_j`. This is the feedstock demand in the allocation. It is a gate (at least 200,000) and a score criterion.
Water (ML/day)	Gate (at least 4) and score criterion.
Grid reliability (0–1)	Fraction of time power is available. Gate (at least 0.7) and score criterion.
Settlement distance (km)	Gate (at least 3) and score criterion.
On protected land (checkbox)	Gate: any protected site is excluded.

Input	Purpose
Protected buffer (km)	Recorded for reference.
Land area (ha)	Gate (at least 50), score criterion and land cost.
Land price (USD/ha)	Score criterion. Land cost is price × area ÷ 20 years.
Tailings suitability (/10)	Gate (at least 5) and score criterion.
Rail distance, substation distance (km)	Score criteria.
Road quality, flatness, labour, environmental sensitivity, seismic risk (/10)	Score criteria.
Fixed opex (USD/yr)	Fixed annual operating cost, part of the total cost.

Criteria (`/criteria`)

Purpose: document the framework and set how much each factor matters.

- The feasibility gates table gives each gate's threshold, definition and required evidence. It is read-only. Thresholds are set in `lib/gates.ts`.
- Criteria are grouped by Industrial, Logistical, Utility, Land and Environmental. Each shows its data field, definition, required evidence and source, direction, range and unit.
- **Weight inputs:** a higher weight gives the criterion more influence on the score. Weights are normalised, so only their relative size matters. Click **Save weights** to store them. Changes apply to the next scenario you run.

Scenario page (`/scenarios/[id]`)

Purpose: the results report. Nothing is entered here.

1. **Recommendation banner:** the minimum-cost site, its score, its annual cost and how many sites passed. It notes when the top-scoring site differs and shows the ROBUST or FRAGILE verdict.
2. **Spatial view:** map of mines, sites and haulage legs. The recommended site is purple, passed sites green, failed sites red and mines amber.
3. **Optimisation model:** the formulation with the run's haulage rate, total supply and supply headroom.
4. **Feasibility gates:** every gate for every site. Hover a badge for the measured value against the threshold.
5. **Weighted scores:** score ranking with a per-criterion breakdown (raw value, normalised 0–100, weighted contribution).
6. **Estimated annual costs:** haulage + fixed opex + land amortisation for each feasible site, ranked by total.

7. **Feedstock allocation:** tonnes, road km and cost per mine for the recommended site.
8. **Sensitivity:**
    - Haulage rate at 0.5×, 0.8×, 1×, 1.2×, 1.5× and 2×: cost and the percent change from the base cost.
    - Mine supply at ±15%: how many sites stay feasible, the optimal site and the cost change.
    - Criterion weights at ±10%, one at a time: whether the top-scoring site changes.
    - **FRAGILE** means the optimal site or top-scoring site changes under a tested perturbation, or the cost-optimal and top-scoring sites differ. Gather better evidence before committing. **ROBUST** means neither changes.

---

5. How a scenario is computed

1. **Gates:** each site is tested against the 7 site gates plus supply sufficiency. Failing any gate excludes it.
2. **Scoring:** for each remaining site, each criterion value is normalised to 0–100 using its min–max range (inverted for cost-type criteria). The score is the weight-averaged sum.
3. **Allocation:** each feasible site's feedstock demand is filled from the nearest mines first, up to each mine's supply.
4. **Cost:** annual cost = haulage + fixed opex + land price × area ÷ 20.
5. **Selection:** the feasible site with the lowest annual cost is recommended.
6. **Sensitivity:** the steps above are repeated under each perturbation.

The *supply-weighted mine distance* criterion is derived by the system. It is the mean road distance to all mines, weighted by their supply, so it needs no input.

6. Data and assumptions

- Data is stored in a local SQLite file, `prisma/dev.db`. Back it up to keep it.
- Distances are straight-line distances times 1.3, not routed roads.
- Cost covers haulage, fixed opex and land amortisation only. Capital cost, processing cost and revenue are not modelled.
- The model selects one plant. The allocation is solved per site, which is exact for a single destination.
- Gate thresholds, the 20-year land amortisation and the 1.3 road factor are set in code (`lib/gates.ts`, `lib/cost.ts`, `lib/geo.ts`).
- Seeded demo values are illustrative and need replacing with surveyed data for a real decision.

7. Project layout

Path	Contents
`app/`	Pages (dashboard, sites, mines, criteria, scenarios) and server actions (`actions.ts`).
`lib/gates.ts`	Gate definitions, thresholds and evaluation.
`lib/scoring.ts`	Normalisation and weighted scoring.
`lib/transport.ts`	Feedstock allocation.
`lib/cost.ts`	Annual cost model.
`lib/evaluate.ts`	Per-site evaluation and cost-optimal selection.
`lib/sensitivity.ts`	Perturbation analysis.
`lib/pipeline.ts`	Scenario assembly.
`lib/demo-data.ts`	Demo sites, mines and criteria with evidence text.
`prisma/`	Database schema and migrations.
`components/`	Map and chart components.

Built with Next.js, React, Prisma (SQLite), Leaflet and Recharts.